

Lecture 1: Examples of Control Problems in Finance

Lecturer: Andrew Papanicolaou

Scribes: Your Name Here

Note: *LaTeX template courtesy of UC Berkeley EECS department.*

Disclaimer: *This note follows Chapter 7.2 of [3] and has not been subjected to formal publication review. Distribution outside class requires instructor permission.*

1.1 The Merton Problem

Following Chapter 7.2 of [3], the portfolio optimization problem of [4, 5] considers an agent who maximizes expected terminal utility,

$$\sup_{\pi \in \mathcal{A}_T} \mathbb{E}U(X_T^\pi), \quad (1.1)$$

subject to

$$\begin{aligned} dS_t &= \mu S_t dt + \sigma S_t dW_t, & S_0 &= s & \text{risky asset} \\ dX_t^\pi &= (\pi_t(\mu - r) + rX_t^\pi) dt + \pi_t \sigma dW_t, & X_0^\pi &= x & \text{agent's wealth} \end{aligned}$$

where:

- W_t is a standard Brownian motion,
- S_t is the risky asset price,
- π_t is the dollar amount invested in the risky asset,
- X_t^π is the wealth process,
- $U(\cdot)$ is the utility function,
- $\mathcal{A}_T$ is the admissible set of square-integrable predictable strategies, $\int_0^T \pi_s^2 ds < \infty$.

A process is **predictable** if, for each time t , its value is known just before time t . A process that is left continuous and adapted to the information at time t , is predictable.

We solve (1.1) to obtain the optimal strategy π . The Merton problem ignores any impact on the price caused by trading. Negligible price impact is an assumption that breaks down for large trades or fast-trading strategies. Price impact and its causes are at the heart of just about any quantitative study of algorithmic trading.

1.2 The Optimal Liquidation Problem

Suppose an agent holds N shares and must liquidate by time T . Market impact implies immediate execution is costly, so liquidation should be distributed over time via a stochastic control problem.

Let ν_t denote the selling rate. The agent chooses a strategy to balance execution cost and inventory risk,

$$\sup_{\nu \in \mathcal{A}_T} \mathbb{E} \left[X_T + Q_T^\nu (S_T - \alpha Q_T^\nu) - \phi \int_0^T (Q_s^\nu)^2 ds \right], \quad (1.2)$$

subject to

$$\begin{aligned} dQ_t^\nu &= -\nu_t dt, & Q_0^\nu &= q && \text{agent's inventory} \\ dS_t^\nu &= -g(\nu_t) dt + \sigma dW_t, & S_0^\nu &= s && \text{asset's fundamental price} \\ \hat{S}_t^\nu &= S_t^\nu - h(\nu_t), & \hat{S}_0^\nu &= s && \text{execution price} \\ dX_t^\nu &= \nu_t \hat{S}_t^\nu dt, & X_0^\nu &= x && \text{agent's cash.} \end{aligned}$$

In the above,

- ν_t is the rate at which agent trades (liquidation rate) is the agent's control,
- Q_t^ν is the agent's inventory,
- W_t a standard Brownian motion,
- S_t^ν is the fundamental price,
- $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is the permanent impact function,
- $\hat{S}_t^\nu$ is the execution price, i.e., the price at which the agents sells,
- X_t^ν is the agent's cash,
- $\mathcal{A}_T$ is the admissible set of bounded, non-negative and predictable strategies; this constraint excludes the repurchasing of shares and keeps the liquidation rate finite.

Remark 1. *The effect from function g is a **reduced-form permanent price impact**, which could be given a macroeconomic interpretation if you embed the model in a larger equilibrium framework.*

1.3 Optimal Limit Order Placement

In the optimal liquidation problem from the previous section consistently be crosses the spread or *walking the book* to sell her shares. Perhaps a better strategy is to place limit orders, which will save the cost of crossing the spread, and perhaps even achieve better performance by posting deeper into the LOB – at a depth of δ_t relative to the midprice S_t ; the risk in doing so is that she may not execute her shares. If $P(\delta_t)$ is the fill probability, the agent solves a control problem over posting depth to optimize expected liquidation value while ensuring full exit by T ,

$$\sup_{\mathcal{A}_T} \mathbb{E} \left[X_T^\delta + Q_T^\delta (S_T^\delta - \alpha Q_T^\delta) - \phi \int_0^T (Q_s^\delta)^2 ds \right], \quad (1.3)$$

subject to

$$\begin{aligned} M_t &= \text{a counting process} && \text{market buy orders} \\ S_t &= S_0 + \sigma W_t && \text{asset midprice} \\ dX_t^\delta &= (S_t + \delta_t)(-dQ_t^\delta), \quad X_0^\delta = x && \text{agent's cash} \\ dQ_t^\delta &= \mathbf{1}\{U_{M_t^{-1}+1} > P(\delta_t)\} dM_t, \quad Q_0^\delta = q && \text{agent's inventory,} \end{aligned}$$

where $U_1, U_2, \dots$ are IID uniform(0,1) random variables, and $M_t^{-1} = \inf\{s > 0 | M_s > t\}$.

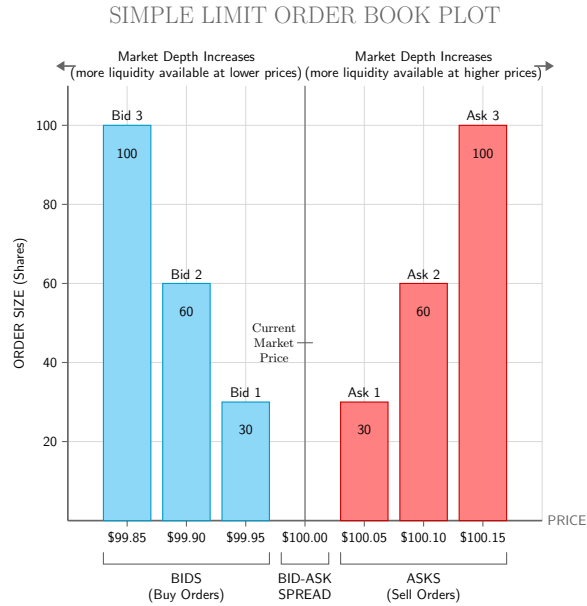


Figure 1.1: An example limit order book. The queues get deeper as you walk farther from the bid-ask, which typically results in price impact being a concave function of trade size.

1.4 Remarks

1. The optimal liquidation problem presented here is originally credited to [1]. We will see later in the semester that it gives rise to optimal TWAP and VWAP execution strategies.
2. The optimal limit order placement problem presented here is based on the work in [2].
3. This lecture was based on Chapter 7.2 in [3]

References

- [1] R. Almgren and N. Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3:5–40, 2001.
- [2] M. Avellaneda and S. Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.
- [3] Á. Cartea, S. Jaimungal, and J. Penalva. *Algorithmic and high-frequency trading*. Cambridge University Press, 2015.
- [4] R. C. Merton. Lifetime portfolio selection under uncertainty: The continuous-time case. *The review of Economics and Statistics*, pages 247–257, 1969.
- [5] R. C. Merton and P. A. Samuelson. *Continuous-time finance*. Blackwell Boston, 1992.